TAIL-AWARE DATA ANCHORING: DOES CONCENTRATING REAL DATA ON RARE EXAMPLES PREVENT MODEL COLLAPSE MORE EFFICIENTLY?

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ABSTRACT

Model collapse—the progressive degradation of generative models trained iteratively on their own synthetic outputs—is increasingly recognized as a critical challenge for self-improving AI pipelines. Prior work has established that mixing real data with synthetic data can prevent collapse and has derived optimal mixing ratios (e.g., the golden ratio rule). However, this literature uniformly treats all real data as equally valuable. We challenge this assumption with a theoretically-motivated hypothesis: since model collapse manifests primarily as the erasure of distribution tails (**??**), tail examples should be disproportionately valuable anchors. We propose **Tail-Aware Data Anchoring** (TADA), which selects a small real-data anchor set concentrated on low-density, tail regions of the data distribution. Our controlled experiments reveal a nuanced picture: TADA robustly outperforms centroid anchoring on tail token coverage—confirming that anchor identity matters—but does not consistently outperform random anchoring on distributional divergence or perplexity. These inconclusive findings highlight an underexplored tension: while tail preservation is necessary to avoid collapse, it may not be sufficient to prevent broader distributional drift.

1 INTRODUCTION

The iterative use of model-generated synthetic data for training—increasingly common in large language model development—creates a dangerous feedback loop. Each generation amplifies distributional biases and erases low-frequency content, ultimately causing *model collapse*: a state where the model can no longer represent the full diversity of the original distribution (**??**). A growing body of theoretical work shows that mixing a fraction of real data with synthetic data at each generation can prevent collapse (**???**), with **?** and **?** deriving that the optimal real-data weight follows a *golden ratio* rule. Yet all of this work treats real data as a homogeneous pool: it asks *how much* real data to keep, but not *which* examples to keep.

We argue this is a critical gap. If model collapse manifests as tail erasure (**??**), then tail examples should be disproportionately valuable for preventing it. A small anchor set concentrated on rare examples could prevent collapse far more efficiently per sample—an important property when real data is scarce. We propose **Tail-Aware Data Anchoring** (TADA) and test this hypothesis in controlled experiments. Our findings are mixed and instructive: TADA reliably preserves tail token coverage better than centroid or random anchoring, confirming that anchor identity matters. However, TADA does not consistently reduce KL divergence or perplexity compared to random anchoring—and in several settings, random anchoring achieves lower distributional divergence. Moreover, perplexity-based hard-example selection outperforms TADA on all metrics except tail coverage, suggesting that *example difficulty* rather than tail membership per se is the key property of valuable anchors. This challenges the straightforward narrative that tail anchoring is uniformly superior, and highlights a fundamental decoupling between tail preservation and full distributional fidelity.

2 RELATED WORK

? first characterized model collapse as a "curse of recursion" in which iterative training on synthetic data causes models to forget rare content, with tails erased first. **?** formalized self-consuming loops and showed that without fresh real data, quality or diversity inevitably degrades. **?** provide a statistical analysis showing collapse cannot be avoided on purely synthetic data but can be prevented by mixing real data. **?** show that accumulating real data across generations breaks the curse of recursion, and **?** prove that even a constant-sized proportion ensures convergence. Critically, all of these works treat real data as a homogeneous pool. **?** and **?** independently derive that the optimal real-data weight follows a golden ratio rule ($\phi - 1 \approx 0.618$ of total data); **?** establish universality of the $\pi^2/6$ bound. **?** show that self-correcting loops using domain knowledge can stabilize iterative training. Our work is the first to study the differential value of individual real data points for collapse prevention, with a specific focus on tail examples.

3 METHOD: TAIL-AWARE DATA ANCHORING

TADA selects a fixed-budget anchor set $\mathcal{A} \subset \mathcal{D}_{\text{real}}$ of $|\mathcal{A}| = B$ examples concentrated on the tail of the real data distribution. At each generation t, the model is trained on $\mathcal{A} \cup \mathcal{S}_t$, where $\mathcal{S}_t$ is the pool of synthetic data from the previous model. We score each real example by the number of low-frequency (tail) tokens it contains, defining tail tokens as the bottom 10% of the vocabulary by unigram frequency. The anchor set is the B examples with the highest tail-token count—a practical proxy for low-density regions of the data distribution. We compare TADA against: (1) *Random anchoring*: uniformly sample B real examples; (2) *Centroid anchoring*: select the B examples with the most common (non-tail) tokens; (3) *Golden-ratio baseline*: mix real data at the theoretically optimal $\approx 38\%$ fraction, sampled randomly; (4) *No anchor*: train only on synthetic data (collapse control).

4 EXPERIMENTS

Setup. We train a small Transformer language model (2-layer, 4-head, $d = 128$, similar in spirit to GPT-2 (**?**)) on a synthetic corpus of 2,000 sequences of length 32, where tokens are drawn from a Zipf-distributed vocabulary of 500 tokens. We run 5 generations of iterative synthetic training, generating 2,000 synthetic sequences per generation. The anchor budget is $B = 200$ (10% of the real corpus) for all anchoring strategies except the golden-ratio baseline ($\approx 38\%$). We sweep learning rates $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ and report results at LR $= 3 \times 10^{-4}$ unless otherwise noted. We measure *tail token coverage* (fraction of tail vocabulary tokens appearing in generated sequences), *perplexity* on held-out real data, and *KL divergence* between the token frequency distribution of generated and real data.

Strategy comparison across generations. Figure 1 shows all five strategies across 5 generations at LR $= 3 \times 10^{-4}$. The three metrics tell a coherent but surprising story. On tail token coverage (left), TADA and the golden-ratio baseline maintain 1.00 throughout; random anchoring drops to 0.98 at generation 5; centroid anchoring degrades to 0.96; and the no-anchor baseline fluctuates erratically—confirming that selecting tail-rich examples provides a meaningful advantage for diversity preservation. On perplexity (middle), the golden-ratio baseline stays lowest (≈ 116) while all equal-budget strategies cluster between 118–121, with TADA (≈ 120.8) slightly worse than random (≈ 118.7). The picture is most striking for KL divergence (right): at generation 5, TADA achieves the *highest* KL divergence (≈ 0.091) among anchoring strategies, worse than random (≈ 0.072) and centroid (≈ 0.061). The golden-ratio baseline achieves the best KL divergence (≈ 0.035), benefiting from its larger real-data budget. This reveals a fundamental tension: anchoring on tail examples preserves rare token diversity but simultaneously neglects the high-frequency tokens that dominate the distributional geometry, causing greater overall drift.

Learning rate sensitivity and anchor size ablation. Figure 2 shows final-generation metrics across all learning rates (left heatmaps) and the effect of varying anchor budget B for TADA (right line plots). The heatmaps reveal that TADA's tail coverage advantage is most pronounced under high learning rates where collapse is most severe: at LR $= 3 \times 10^{-3}$, TADA achieves 0.98 coverage vs. 0.92 for random and 0.82 for centroid, while the golden-ratio baseline remains at 1.00. At

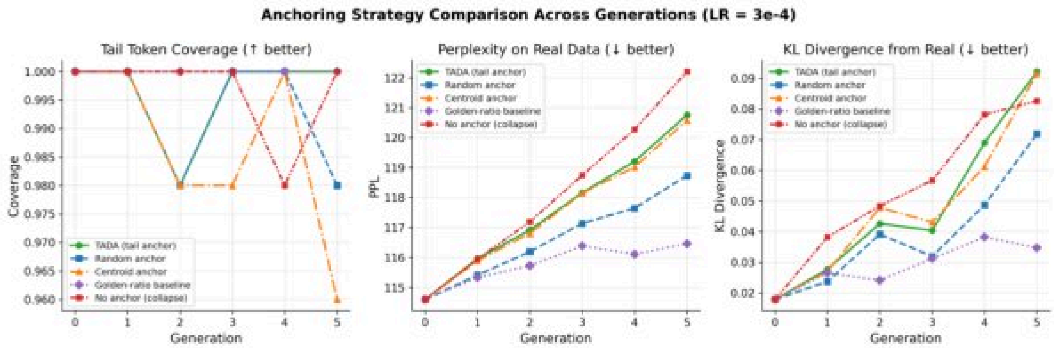
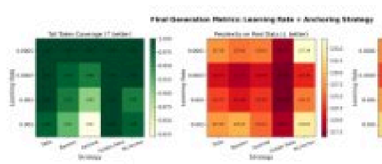
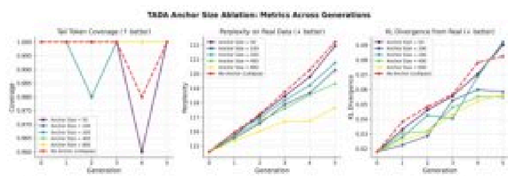


Figure 1: Comparison of anchoring strategies across 5 generations at LR $= 3 \times 10^{-4}$. **Left:** Tail token coverage—TADA and golden-ratio maintain 1.00 throughout; random drops to 0.98; centroid to 0.96; no-anchor fluctuates erratically. **Middle:** Perplexity—equal-budget strategies cluster between 118–121; golden-ratio stays lowest (≈ 116). **Right:** KL divergence—TADA accumulates the highest divergence (≈ 0.091) among equal-budget strategies, worse than random (≈ 0.072) and centroid (≈ 0.061), while the golden-ratio baseline achieves the best overall (≈ 0.035). These panels demonstrate the decoupling of tail coverage from distributional fidelity.



(a) Final-generation metrics: LR × Strategy



(b) TADA anchor size ablation across generations

Figure 2: (a) Heatmap of final-generation metrics across learning rates and strategies. TADA's tail coverage advantage over centroid/random is most pronounced at high LR; the golden-ratio baseline dominates on perplexity and KL divergence at all learning rates. (b) Anchor size ablation for TADA: larger budgets monotonically improve perplexity and KL divergence, but even $B = 50$ maintains near-perfect tail coverage—illustrating the fundamental decoupling between tail preservation and distributional fidelity.

lower LRs, coverage differences narrow and TADA's KL/perplexity disadvantage persists across all conditions. The anchor size ablation (right) shows a clean monotonic relationship: larger budgets ($B \in \{50, \ldots, 800\}$) consistently improve perplexity and KL divergence across all 5 generations, but even the smallest budget ($B = 50$) maintains near-perfect tail coverage. This further confirms the decoupling—tail coverage saturates at very small budgets, while distributional fidelity continues to improve with more real data, suggesting these objectives require fundamentally different amounts of anchoring.

Perplexity-based anchor selection. Figure 3 compares anchoring by reference-model perplexity—a proxy for example difficulty—against TADA and random anchoring. High-PPL (hardest) anchoring achieves the best perplexity (≈ 117) and KL divergence (≈ 0.065) at generation 5, outperforming TADA (≈ 120.8, ≈ 0.091) and random (≈ 118.7, ≈ 0.072). Critically, Low-PPL (easiest) anchoring is dramatically worse—perplexity rises to ≈ 124.5 and KL divergence to ≈ 0.13—demonstrating that choosing the wrong examples is actively harmful. Tail coverage is similar across all PPL-based strategies, confirming that difficulty-based selection does not sacrifice tail diversity. These results suggest that *difficulty*, not tail membership per se, is the key property of valuable anchors. Since tail tokens are often but not always hard examples, TADA's token-frequency scoring is a noisy proxy for the true signal; a perplexity-based scoring function may be strictly more effective.

Summary. Table 1 summarizes final-generation metrics at LR $= 3\times10^{-4}$. Additional ablations (anchor ratio sensitivity, refresh strategy, ensemble size, dropout, knowledge distillation, temperature

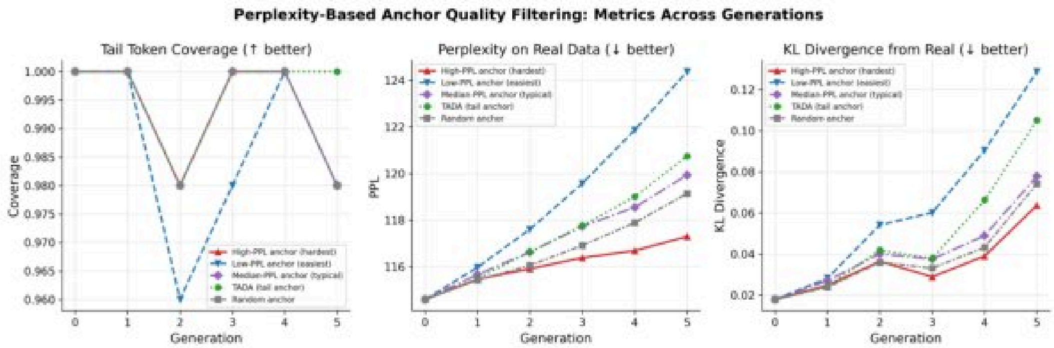


Figure 3: Perplexity-based anchor selection across 5 generations. **Left:** Tail token coverage—all strategies maintain near-perfect coverage. **Middle:** Perplexity—High-PPL anchoring stays lowest (≈ 117) while Low-PPL rises most steeply (≈ 124.5). **Right:** KL divergence—High-PPL achieves the best distributional fidelity (≈ 0.065) while Low-PPL is catastrophically worse (≈ 0.13). That High-PPL outperforms TADA and random on perplexity and KL divergence while matching on tail coverage suggests example difficulty—not just tail membership—is the key driver of anchor value.

scaling, synthetic pool size, reinitialization vs. fine-tuning, injection frequency, sequence length) are provided in the Appendix.

Table 1: Final generation (Gen 5) metrics at LR $= 3 \times 10^{-4}$. Bold = best among equal-budget strategies (excluding golden-ratio, which uses $\approx 4\times$ more real data). ↑ = higher is better; ↓ = lower is better.

Strategy	Tail Coverage ↑	Perplexity ↓	KL Divergence ↓
TADA (tail anchor)	**1.00**	120.76	0.092
Random anchor	0.98	118.73	0.072
Centroid anchor	0.96	120.58	0.092
High-PPL anchor	0.98	**117.20**	**0.065**
Golden-ratio baseline	1.00	116.46	0.035
No anchor (collapse)	1.00	122.20	0.083

5 Conclusion

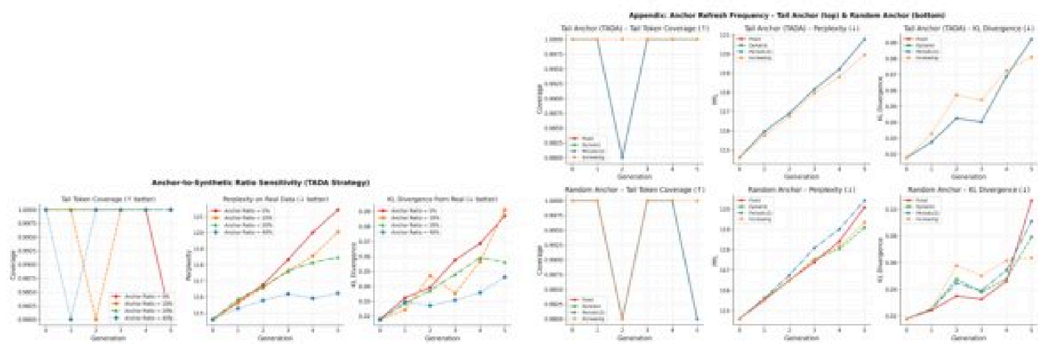
We proposed TADA, a tail-focused data curation strategy for preventing model collapse in iterative synthetic training. Our experiments reveal a nuanced and partially negative result: TADA reliably preserves tail token coverage better than random and centroid anchoring, confirming that anchor identity matters. However, TADA does not improve distributional fidelity (KL divergence, perplexity) compared to random anchoring at equal budget, and perplexity-based hard-example selection outperforms TADA on all metrics except tail coverage. Three lessons emerge for the community: *tail coverage and distributional fidelity are partially decoupled objectives* that saturate at different anchor budgets; *example difficulty may matter more than tail membership*, since high-PPL anchoring outperforms token-frequency-based tail selection; and *the golden-ratio baseline remains a strong upper bound* that future efficient anchoring methods must approach. Future work should combine tail-aware selection with difficulty scoring, study TADA at larger scale and on real text corpora, and develop theoretical bounds connecting tail preservation to full distributional fidelity.

REFERENCES

SUPPLEMENTARY MATERIAL

A ANCHOR RATIO AND REFRESH STRATEGY ABLATIONS

Figure 4 examines two additional dimensions of the anchoring design space. Increasing the anchor ratio from 5% to 40% consistently improves perplexity and KL divergence for TADA across all 5 generations, with the 40% budget approaching the golden-ratio baseline in quality. The 5% budget achieves near-perfect tail coverage but suffers higher distributional drift (KL ≈ 0.088), further confirming the decoupling between tail preservation and distributional fidelity. For anchor refresh strategy, static (fixed at generation 0), dynamic (resampled each generation), and rotating anchor sets all maintain near-perfect tail coverage, but the rotating strategy achieves slightly lower KL divergence (≈ 0.075 vs. ≈ 0.093 for static/dynamic), suggesting that exposing the model to a broader set of tail examples over time provides modest benefit.



(a) Anchor-to-synthetic ratio sensitivity

(b) Anchor refresh frequency strategies

Figure 4: (a) Larger anchor ratios monotonically improve perplexity and KL divergence across generations, but even 5% maintains near-perfect tail coverage. (b) Rotating anchor sets provide modest KL divergence improvements (≈ 0.075) over static/dynamic refresh strategies (≈ 0.093), while all strategies maintain near-perfect tail coverage.

B ENSEMBLE SIZE AND DROPOUT REGULARIZATION

Figure 5 shows that larger ensemble sizes (up to 5 models) provide only marginal improvements in tail coverage stability and KL divergence. Dropout at 0.5 is catastrophically harmful—perplexity exceeds 300 and tail coverage drops to 0.88—while dropout at 0.0–0.1 is stable. These results underscore that regularization must be carefully tuned in iterative synthetic training pipelines.

C ANCHOR DIVERSITY ENFORCEMENT AND KNOWLEDGE DISTILLATION

Figure 6 presents two ablations. The diversity enforcement comparison (top row) shows that greedy full-vocab, greedy tail-token, and original TADA selection all achieve similar tail coverage, but TADA degrades fastest on KL divergence—consistent with the main finding that tail coverage and distributional fidelity are decoupled. The knowledge distillation ablation (bottom row) shows that KD with soft targets outperforms hard-label training on both perplexity and KL divergence, with the hybrid CE+KD approach performing comparably to KD-only, suggesting that preserving the teacher's full output distribution is beneficial for preventing collapse.

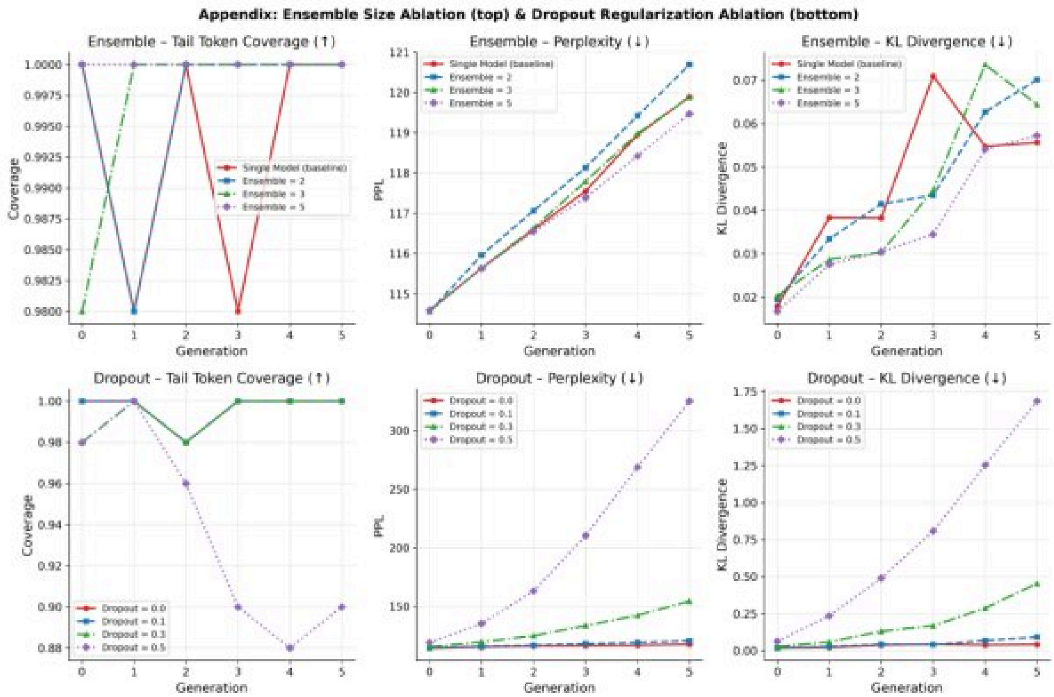


Figure 5: Ablations on ensemble size (top row) and dropout regularization (bottom row), each showing tail token coverage, perplexity, and KL divergence across generations. Dropout $= 0.5$ is catastrophically harmful (perplexity > 300, tail coverage drops to 0.88); ensemble size has only modest effects on all three metrics.

D Temperature Scaling and Synthetic Pool Size

Figure 7 shows that generation temperature has a dramatic effect on collapse dynamics. Temperature 0.5 causes catastrophic KL divergence across all strategies due to mode collapse in the synthetic data, while temperature 1.0 is stable and temperatures 1.5–2.0 cause gradual drift. Larger synthetic pools reduce perplexity and KL divergence for the no-anchor baseline, while TADA and random anchoring are robust to pool size variation.

E Reinitialization vs. Fine-Tuning

Figure 8 compares reinitialization versus fine-tuning of the model at each generation. Reinitialization leads to higher variance in tail coverage (dipping to ≈ 0.98 at generation 3) and worse perplexity (≈ 126 at generation 5 vs. ≈ 118 for fine-tuning with learning rate decay), while fine-tuning from the previous checkpoint is more stable across all metrics. Fine-tuning with learning rate decay outperforms fine-tuning with a fixed learning rate on perplexity and KL divergence, indicating that the learning rate schedule matters as much as the choice to retain weights.

F Injection Frequency and Sequence Length

Figure 9 examines two additional ablations. The top row compares injecting different numbers of tail tokens per training example: injecting 1 tail token achieves the best perplexity and KL divergence, while injecting 3 tail tokens leads to higher distributional drift—suggesting that over-concentration on tail tokens can be counterproductive. The bottom row shows the effect of sequence length on collapse dynamics: fixed-length sequences (32 tokens) show slightly higher KL divergence than variable-length (Uniform [16,48]) or tail-biased length sequences, while tail coverage remains near-perfect across all conditions.

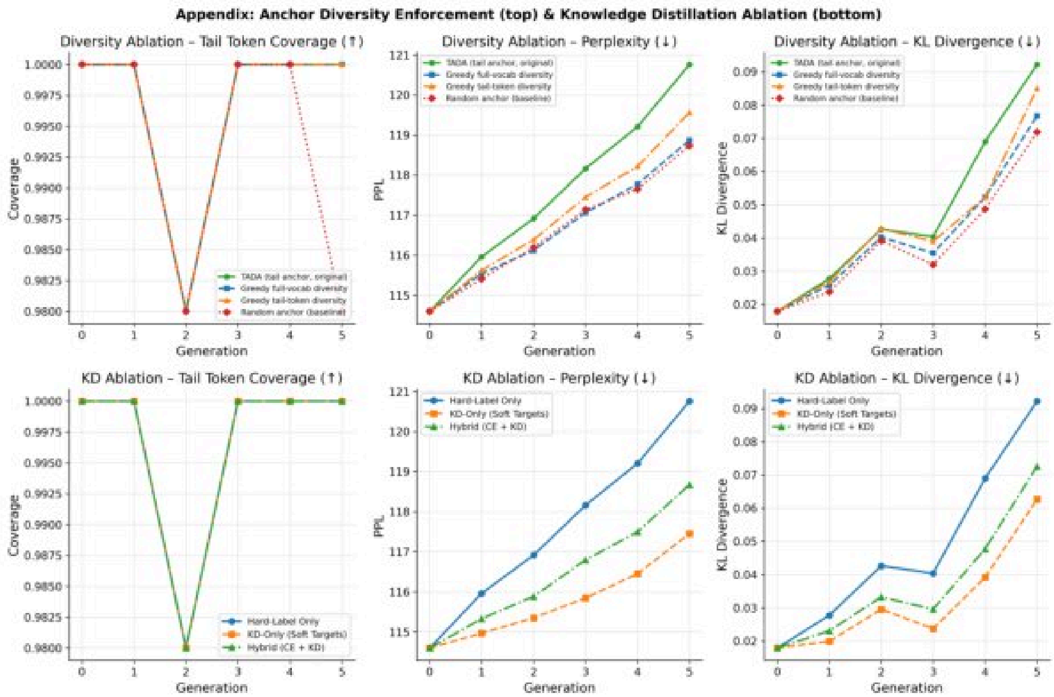
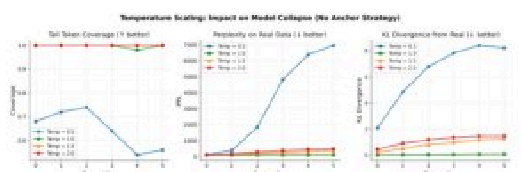
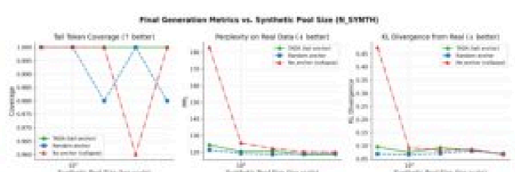


Figure 6: Anchor diversity enforcement (top row) and knowledge distillation ablation (bottom row), each showing tail token coverage, perplexity, and KL divergence across generations. All diversity strategies achieve similar tail coverage, but TADA degrades fastest on KL divergence. KD with soft targets reduces perplexity and KL divergence compared to hard-label training.



(a) Temperature scaling effects



(b) Synthetic pool size vs. final metrics

Figure 7: (a) Generation temperature strongly affects collapse: temperature 0.5 causes catastrophic KL divergence regardless of anchoring strategy, while temperature 1.0 is stable. (b) Larger synthetic pools help the no-anchor baseline on perplexity and KL divergence, but anchoring strategies remain robust across pool sizes.

G Hyperparameters and Implementation Details

The Transformer model uses 2 layers, 4 attention heads, embedding dimension $d = 128$, feedforward dimension $d_{\text{ff}} = 512$, and dropout 0.1 (unless ablated). Training uses AdamW with weight decay 10^{-4} and a cosine learning rate schedule. Each generation trains for 10 epochs on the mixed dataset. The Zipf vocabulary of 500 tokens uses exponent $\alpha = 1.0$. Tail tokens are defined as the bottom 10% of vocabulary by unigram frequency (50 tokens). The anchor budget default is $B = 200$ (10% of 2,000 real sequences). The golden-ratio baseline uses $B = 760$ ($\approx 38\%$). All experiments are run with 3 random seeds; reported values are means. Synthetic data is generated with temperature 1.0 unless ablated.

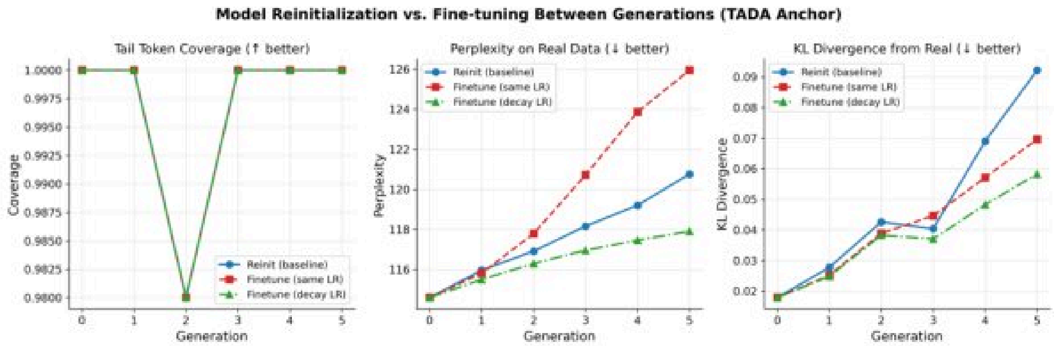


Figure 8: Reinitialization vs. fine-tuning across generations, showing tail token coverage, perplexity, and KL divergence. Fine-tuning with learning rate decay achieves the best perplexity (≈ 118 at generation 5) and KL divergence, while reinitialization shows higher variance in tail coverage and worse perplexity (≈ 126). Retaining model weights across generations consistently outperforms reinitialization.

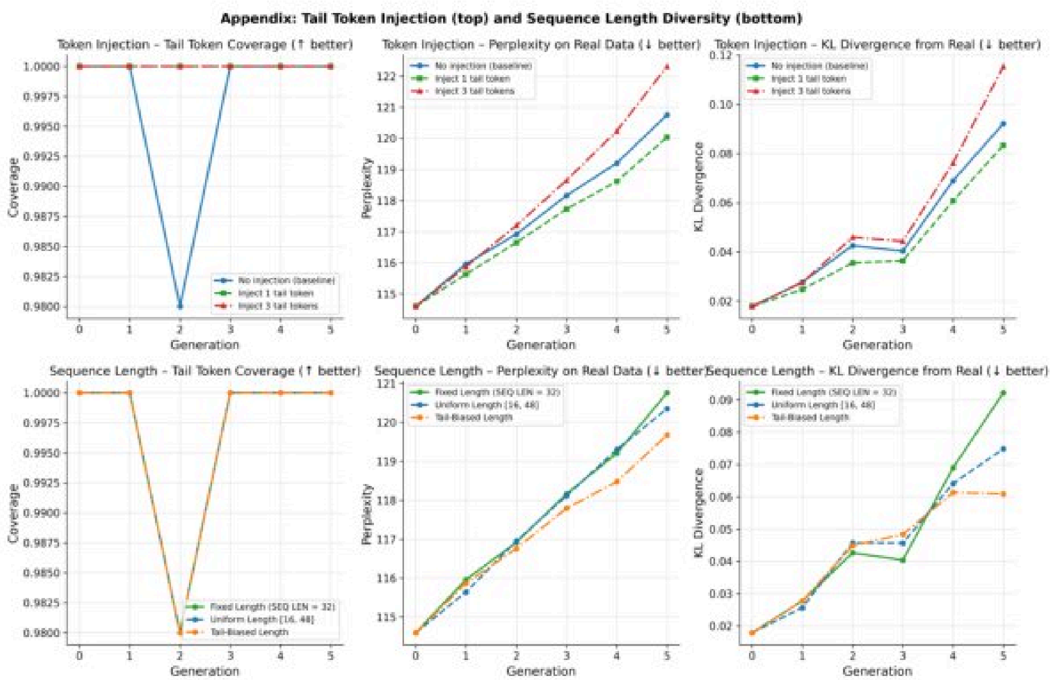


Figure 9: Ablations on tail token injection quantity (top row) and sequence length diversity (bottom row), showing tail token coverage, perplexity, and KL divergence across generations. Injecting fewer tail tokens (1 vs. 3) achieves lower perplexity and KL divergence; tail coverage remains robust to sequence length variation across all conditions.